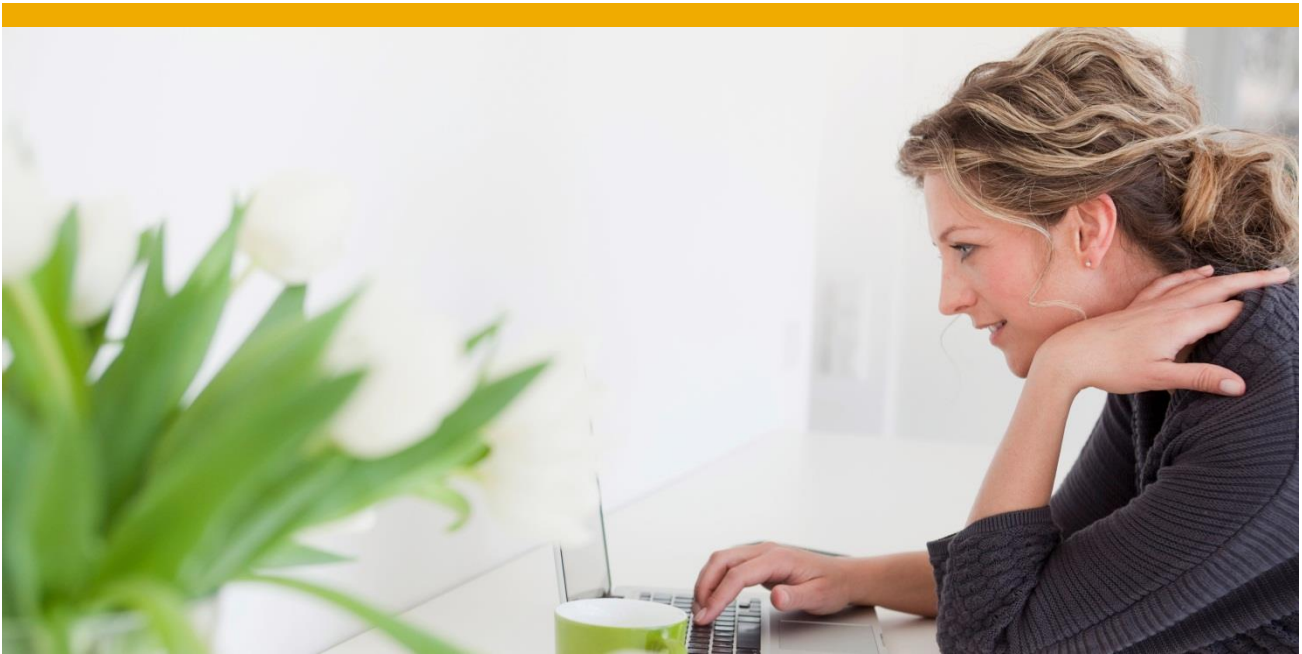




How to Configure Fiori Launchpad using Azure Identity Provider Step-by-Step

SAP Netweaver or S4H – Gateway

Ali Chalhoub



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Abstract

Chapter 1 - Configuring SAP Service Provider

- 1.1. Overview of the Architecture
- 1.2. Configuring Scenario Service Provider
- 1.3. Configuring Scenario Identity Provider
- 1.4. Downloading Identity Provider Metadata
- 1.5. Importing Azure Identity Provider Certificate into SAP S/4HANA Service Provider

Chapter 2 - Configuring Fiori Launchpad to Support SAML2

- 2.1. Configuring Fiori Launchpad
- 2.2. Configuring IDP to support Login Name
- 2.3. Testing SAML Using Fiori launchpad
- 2.4. Configuring Single Logout Endpoint
- 2.5. Configuring Fiori Launchpad Designer

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- 5.1. No RelayState mapping found for RelayState value
- 5.2. How to trace SAML issues in S/4HANA or Netweaver
- 5.3. Microsoft Tutorial on SAML and Netweaver using Azure

Chapter 1

Configuring SAP Service Provider

Welcome to How to Configure Fiori Launchpad to Support SAML2 Using Microsoft Azure Identity Provider Step-by-Step. In this e-book you will find all the details are needed to let you configure a Fiori launchpad on on-premise SAP S/4HANA 1909 system. In this eBook we will discuss and show the user how to configure:

- Azure Active Directory Single sign-on
- Fiori launchpad on on-premise system running S/4HANA 1909 or higher

Note

To make the process simple, the steps provided in this book are done against a single NetWeaver Gateway system no ERP involved.

Disclaimer

Any errors or configuration issues or support issues regarding Microsoft Azure Active Directory Single Sign-on, it should be followed with Microsoft Support and not SAP Product Support. SAP is not responsible for any issues or support issues related to Microsoft Azure.

Requirement

1. NetWeaver 7.5 or higher
2. Fiori launchpad already configured and working with SSL support
3. Administrator has an account with Microsoft Azure that has access to creating application from a gallery

Overview of the Architecture

Before we can start our configuration, we need to look at the Architecture that this book will address. This eBook will cover the following scenario:

Microsoft Azure Active Directory Single sign-on with SAP Fiori launchpad running on on-premise S/4HANA system.

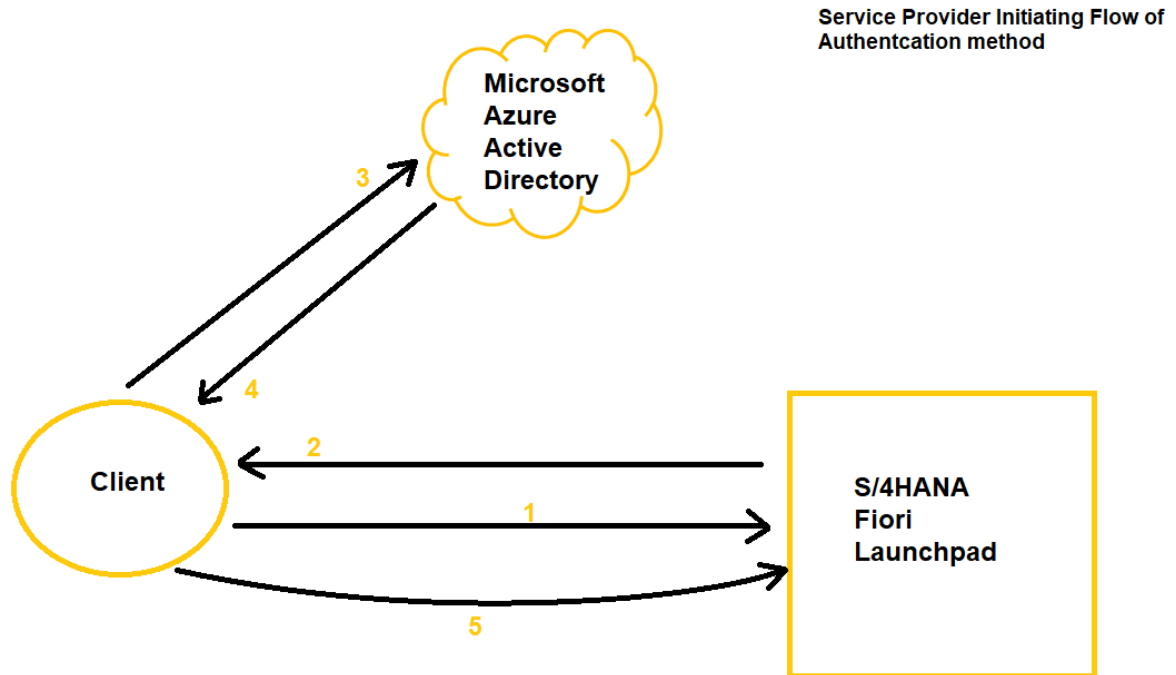


Figure 1 Microsoft Azure Active Directory with Fiori launchpad

1. A web client makes a request to SAP Fiori launchpad
2. SAP Fiori launchpad (SP) redirects the client to Microsoft Azure Active Directory
3. Client is asked to authenticate with Microsoft Azure Active Directory
4. After the client is authenticated successfully, a SAML XML assertion is generated which contains all the information needed about the client such as user id, first name, last name and all this sent to the client
5. The client makes a post request to SAP Fiori launchpad where the XML assertion is validated at the NetWeaver level and a session is created and the client is granted access to Fiori launchpad

Parameters Configuration in S/4HANA System

- In order to configure SAML in S/4HANA, the following services http and https must be active and functioning
- The following profile parameters should match what it is shown below:

Current Values of Relevant Profile Parameters:

```
login/create_sso2_ticket = 2
login/accept_sso2_ticket = 1
login/ticketcache_entries_max = 1000
login/ticketcache_off = 0
login/ticket_only_by_https = 0
icf/set_HTTPOnly_flag_on_cookies = 3
icf/user_recheck = 0
http/security_session_timeout = 1800
http/security_context_cache_size = 2500
rdisp/plugin_auto_logout = 1800
rdisp/autothtime = 60
```

Service /sap/public/bc/icf/logoff is active
Service /sap/public/myssocntl is active

Activating required SICF services

In this section we need to activate few SICF services. Here is the list

```
/sap/public/bc/sec/cdc_ext_service
/sap/public/bc/sec/saml2
/sap/bc/webdynpro/sap/saml2
/sap/bc/webdynpro/sap/sec_diag_tool
```

To activate the above services, we will show you the process of how to do it for one and repeat the process for the rest. For example, to activate /sap/public/bc/sec/cdc_ext_service, do the following:

1. Login to S/4HANA system
2. Issue the following tCode SICF

3. Enter the service path of /sap/public/bc/sec/cdc_ext_service as shown below:

 < Create Host/Service     → External Aliases   System Monitor Active  More ▾

Filter Details

Virtual Host:  Service Path:

ServiceName:

Description:

Lang.: Reference:

 ▾

Virtual Hosts/Services	Documentation	Reference Service
▾  default_host	VIRTUAL DEFAULT HOST	
▾  sap	SAP NAMESPACE; SAP IS OBLIGED NOT TO DELIVER ANY...	
▾  public	PUBLIC SERVICES	
▾  bc	Basis Tree (Basis Functions)	
▾  sec	Security Node	
 cdc_ext_service	Common Domain Cookie external service	

4. If it is grayed out, right click on cdc_ext_service and select Activate Service

The screenshot displays the SAP Service Manager interface. At the top, there is a toolbar with icons for creating, deleting, refreshing, and viewing external aliases. Below this is the 'Filter Details' section, which includes input fields for 'Virtual Host', 'Service Path' (set to '/sap/public/bc/sec/cdc_ext_service'), 'ServiceName', 'Description', 'Lang.' (set to 'English'), and 'Reference'. There are buttons for 'Apply', 'Reset', and 'Fine-Tune'.

Below the filter details is a table with three columns: 'Virtual Hosts/Services', 'Documentation', and 'Reference'. The table lists the following entries:

Virtual Hosts/Services	Documentation	Reference
default_host	VIRTUAL DEFAULT HOST	
sap	SAP NAMESPACE; SAP IS OBLIGED NOT TO DELIVER ANY...	
public	PUBLIC SERVICES	
bc	Basis Tree (Basis Functions)	
sec	Security Node	
cdc_ext_service	Common Domain Cookie external service	

A right-click context menu is open over the 'cdc_ext_service' entry. The menu options are:

- New Subelement
- Display Service
- Delete Service
- Rename Service
- Activate Service**
- Deactivate Service
- Test Service
- References to Service
- Obj. Directory Entry
- Cut
- Copy
- Paste

5. Select Yes as shown below:

The screenshot shows the SAP Fiori Launchpad configuration interface. At the top, there is a toolbar with buttons like 'Create Host/Service', 'External Aliases', and 'System Monitor Active'. Below the toolbar is the 'Filter Details' section with fields for 'Virtual Host', 'Service Path' (set to '/sap/public/bc/sec/cdc_ext_service'), 'ServiceName', 'Description', 'Lang.' (set to 'English'), and 'Reference'. An 'Apply' button is visible. On the left, a tree view shows the hierarchy: 'Virtual Hosts/Services' > 'default_host' > 'sap' > 'public' > 'bc' > 'sec' > 'cdc_ext_service'. The 'cdc_ext_service' is highlighted. A dialog box titled 'Activation of ICF Services' is open in the foreground. It asks 'Do you want to activate service /default_host/sap/public/bc/sec/cdc_ext_service?'. There are four buttons: 'Yes', 'Yes' (highlighted in yellow), 'Info', and 'Cancel'.

6. Repeat the same process for the rest of the services

Configuring Local Provider

In this scenario we will be configuring SAP Fiori launchpad on-premise to authenticate with Microsoft Azure Active Directory.

Note

In this section there is an assumption, that Fiori launchpad is configured in S/4HANA system and Fiori launchpad can be accessed using HTTPS. As well the user does have access to Microsoft Azure Active Directory.

2. Connecting to SAP Service Provider. In our configuration that would be our S/4HANA System

1. Login to SAP S/4HANA System
2. Execute tCode **saml2** or execute this from the browser
`https://<HOST-NAME>:<PORT>/sap/bc/webdynpro/sap/saml2?sap-client=<CLIENT-ID>`

Note

It is important the URL in the browser after execution of saml2, must be the full domain name including the port if it is not port 443. The URL domain must be reachable externally. That could be the Web Dispatcher or the fully qualified name. (i.e If you can access the configuration page using http after executing saml2, then the URL must be changed manually

to include the fully qualified domain and SSL port if it is not 443 and https protocol.

3. Login as shown below

The image shows the SAP NetWeaver login interface. On the left, there is a photograph of a man in a dark sweater leaning over a desk, writing on a document. To the right of the photo, the text "SAP NetWeaver" is displayed in a bold, orange font. Below this, there are input fields for "System:", "Client: *", "User: *", "Password: *", and a "Language:" dropdown menu set to "English". A yellow "Log On" button is positioned below the language dropdown. Underneath the login fields, there is a blue link that says "Change Password". At the bottom of the interface, there is a large, empty rectangular box with a black border and a vertical scrollbar on the right side. In the bottom right corner, the SAP logo is visible, and the text "Copyright © 2020 SAP SE. All rights reserved." is displayed.

4. Click on Enable SAML 2.0 Support if no SAML has been configured in the system
5. We should see the following screen below

The image shows the "SAML 2.0 Configuration" screen. At the top, the title "SAML 2.0 Configuration" is displayed. Below the title, a message states "Client [redacted] is not configured to support SAML 2.0". Underneath this message, there are three buttons: "Enable SAML 2.0 Support", "Create SAML 2.0 Local Provider", and "Import SAML 2.0 Configuration".

Figure 4 Enabling SAML2 in S/4HANA

6. Select **Create SAML 2.0 Local Provider**
7. Now enter a name that represent the Local Provider Configuration. Azure requirement of the name must be <protocol>://<NAME>, in our case that would be <protocol>://<sid><client> (i.e

https://<sid><client>)

SAML 2.0 Local Provider Configuration

1

2

3

Initial SettingsGeneral SettingsService Provider Settings

< Previous

Next >

Finish

Cancel

* Provider Name:

Operation Mode:

Service Provider

< Previous

Next >

Finish

Cancel

Figure 5 Providing name to the Local Provider Service Provider

8. Click **Next**

9. On this screen below do not do anything, click **Next** as well

SAML 2.0 Local Provider Configuration

1 Initial Settings 2 **General Settings** 3 Service Provider Settings

< Previous Next > Finish Cancel

Miscellaneous

Clock Skew Tolerance: Seconds

< Previous Next > Finish Cancel

Figure 6 Miscellaneous

10. Under Identity Provider Discovery: Common Domain Cookie (CDC), make sure selection Mode is set to **Automatic** as shown below:

SAML 2.0 Local Provider Configuration

1 Initial Settings 2 General Settings 3 **Service Provider Settings**

< Previous Next > Finish Cancel

Identity Provider Discovery: Common Domain Cookie (CDC)

Selection Mode:

Miscellaneous

Affiliation Name:

Assertion Consumer Service

Supported Bindings: ☒ HTTP POST ☒ HTTP Artifact ☒ PAOS

Single Logout Service

Supported Bindings: ☒ HTTP Redirect ☒ HTTP POST ☒ HTTP Artifact ☒ SOAP

Artifact Resolution Service

Mode:

Artifact Validity Period: Seconds

Figure 7 Setting selection Mode

Note

Selection Mode Automatic means the user will not need to select the default authentication provider. It will be selected automatically.

11. Click **Finish**

12. We should see the following screen below:

SAML 2.0 Local Provider Configuration

The screenshot shows the 'SAML 2.0 Local Provider Configuration' wizard at step 3, 'Service Provider Settings'. The progress bar at the top indicates three steps: 1. Initial Settings, 2. General Settings, and 3. Service Provider Settings (current). Below the progress bar are navigation buttons: '< Previous', 'Next >', 'Finish', and 'Cancel'. The main configuration area is divided into two columns. The left column contains 'Identity Provider Discovery: Common Domain Cookie (CDC)' with a 'Selection Mode' dropdown set to 'Automatic', and 'Miscellaneous' with an 'Affiliation Name' text field. The right column contains 'Assertion Consumer Service' with 'Supported Bindings' (HTTP POST, HTTP Artifact, PAOS) all checked; 'Single Logout Service' with 'Supported Bindings' (HTTP Redirect, HTTP POST, HTTP Artifact, SOAP) all checked; and 'Artifact Resolution Service' with 'Mode' set to 'Enabled' and 'Artifact Validity Period' set to '60' seconds. At the bottom of the configuration area are the same navigation buttons: '< Previous', 'Next >', 'Finish', and 'Cancel'.

Figure 8 Creating Local Provider Configuration

13. Next, we need to download the Metadata of our Local Provider, so it can be imported into Azure Active Directory single sign-on configuration. Click on Metadata as shown below:

SAML 2.0 Configuration of SAP System: [redacted]

[Local Provider](#)

[Trusted Provider](#)

[Policies](#)

[Name ID Management](#)

[Edit](#)

[Save](#)

[Cancel](#)

[Disable](#)

[Metadata](#)

[Delete Configuration](#)

[Export Configuration](#)

Provider Name:

Operation Mode:

Service Provider

Status:

☒ Enabled

[General Settings](#)

[Authentication Contexts](#)

[Service Provider Settings](#)

Signature and Encryption

Signing Keypair:

[Details](#)

Encryption Keypair:

[Details](#)

☐ Include Certificate in Signature

☒ Sign Metadata

Miscellaneous

Clock Skew Tolerance:

120

Seconds

Figure 9 Accessing Metadata information

14. Click on Download Metadata

SAML 2.0 Metadata

Choose which entity descriptor to be included in the metadata

☒ Service Provider

☒ Application Service Provider

☒ Security Token Service

Download Metadata

Close

Figure 10 Downloading Metadata xml information

15. Save the XML file to your local machine because it will be required in the next step when we configure the Identity Provider. We are going to call the xml file s4hana_host.xml

Configuring Azure Active Directory Single sign-on

3. Connecting to Microsoft Azure Active Directory Single sign-on if it is not already configured

1. Open your Web browser
2. Enter the URL of the Microsoft Azure Identity Provider. For example:
IDP Host: <https://portal.azure.com/?quickstart=True#home>
3. Once logged in the screen may look like the one below. Click on Azure Active Directory

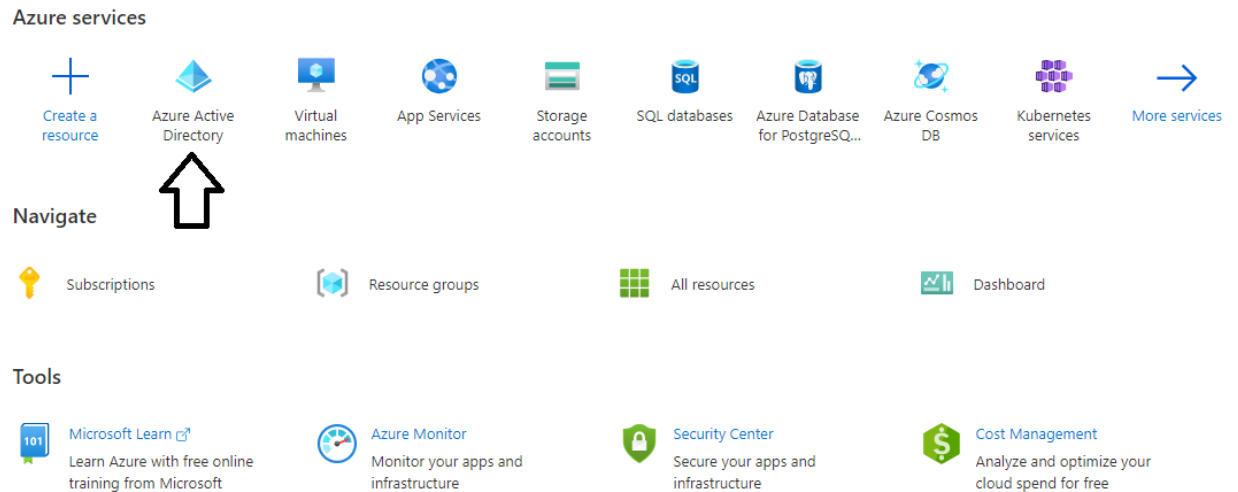


Figure 11 Microsoft Azure Services

4. Click on Enterprise applications

The screenshot shows the Microsoft Azure portal interface for the Default Directory. The top navigation bar includes the Microsoft Azure logo and a search bar. Below the navigation bar, the breadcrumb trail shows 'Home > Default Directory'. The main heading is 'Default Directory | Overview' with a subheading 'Azure Active Directory'. The left sidebar contains a list of navigation items: Overview, Getting started, Preview hub, Diagnose and solve problems, Manage (Users, Groups, External Identities, Roles and administrators, Administrative units, Enterprise applications, Devices, App registrations, Identity Governance, Application proxy, Licenses, Azure AD Connect, Custom domain names, Mobility (MDM and MAM), Password reset, Company branding, User settings, Properties, Security), and Monitoring. The 'Enterprise applications' link is highlighted in yellow. The main content area shows the 'Default Directory' overview with a search bar and two sections: 'Tenant information' and 'Azure AD Connect'. The 'Tenant information' section displays 'Your role: Global administrator' and a link to 'More info'. The 'Azure AD Connect' section shows 'Status: Not enabled' and 'Last sync: Sync has never run'. Below these sections is a 'Sign-ins' section with a bar chart showing '1' sign-in and a date 'Sep 13'. At the bottom, there is a 'Create' section with icons for User, Guest user, Group, Enterprise application, and App registration.

Figure 12 Accessing Applications section

5. Click on + New application

The screenshot shows the Microsoft Azure portal interface. At the top, the navigation bar includes the Microsoft Azure logo, a search bar, and several icons. Below the navigation bar, the breadcrumb trail reads 'Home > Default Directory > Enterprise applications'. The main heading is 'Enterprise applications | All applications', with a subheading 'Default Directory - Azure Active Directory'. On the left, there is a sidebar with various navigation options: Overview, Manage, Security, Activity, and Troubleshooting + Support. The 'Manage' section is expanded, showing 'All applications' as the selected option. In the main content area, there is a '+ New application' button, a 'Columns' button, a 'Preview features' button, and a 'Got feedback?' link. Below these buttons, there is a notification banner about the 'Enterprise Apps search preview'. Underneath the banner, there are three dropdown menus: 'Application type' (set to 'Enterprise Applications'), 'Applications status' (set to 'Any'), and 'Application visibility' (set to 'Any'). To the right of these dropdowns are 'Apply' and 'Reset' buttons. Below the filters, there is a search bar with the text 'First 50 shown, to search all of your applications, enter a display name or the application ID.' and a table with columns: Name, Homepage URL, Object ID, and Application ID. The table currently displays the message 'Didn't find what you're looking for? Click 'Add' above to add a new application.'

Figure 13 Adding new application

6. Search for NetWeaver application from the gallery as shown below

[Home](#) > [Default Directory](#) > [Enterprise applications](#) >

Browse Azure AD Gallery (Preview)

[+ Create your own application](#) [Request new gallery app](#) | [Got feedback?](#)

[Click here to switch back to the old app gallery experience. →](#)

Single Sign-on : **All**

User Account Management : **All**

Categories : **All**

[Federated SSO](#) [Provisioning](#)

Showing 2 of 2 results



Figure 14 Creating application

7. Once it is found, click on it. We should see the screen below

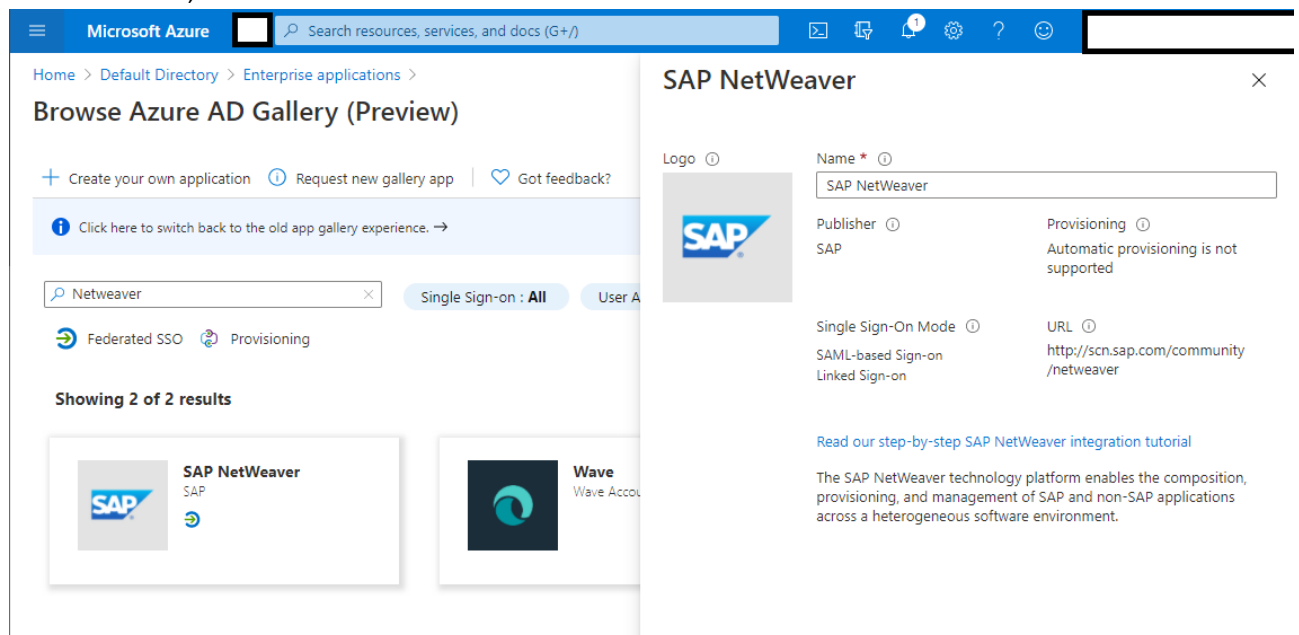


Figure 15 Configuration screen of the application

8. Click on Create
9. We should see the following screen below

The screenshot displays the Microsoft Azure portal interface for the SAP NetWeaver application. At the top, the navigation bar shows the Microsoft Azure logo, a search bar, and various utility icons. Below the navigation bar, the breadcrumb trail reads: Home > Default Directory > Enterprise applications > Browse Azure AD Gallery (Preview) > SAP NetWeaver. The main header area features the SAP NetWeaver logo and the title 'Overview' with a close button (X) on the right.

The left-hand navigation pane is organized into several sections: Overview (selected), Deployment Plan, Diagnose and solve problems, Manage, Properties, Owners, Roles and administrators (Preview), Users and groups, Single sign-on, Provisioning, Self-service, Security, Conditional Access, Permissions, Token encryption, Activity, Sign-ins, Usage & insights (Preview), Audit logs, Provisioning logs (Preview), Access reviews, Troubleshooting + Support, Virtual assistant (Preview), and New support request.

The main content area is divided into three primary sections:

- Properties:** This section contains three input fields: 'Name' (pre-filled with 'SAP NetWeaver'), 'Application ID' (empty), and 'Object ID' (empty). Each field has a small icon to its right.
- Getting Started:** This section contains five numbered steps, each with an icon, a title, a description, and a link:
 - 1. Assign users and groups:** Provide specific users and groups access to the applications. Link: [Assign users and groups](#)
 - 2. Set up single sign on:** Enable users to sign into their application using their Azure AD credentials. Link: [Get started](#)
 - 3. Provision User Accounts:** You'll need to create user accounts in the application. Link: [Learn more](#)
 - 4. Conditional Access:** Secure access to this application with a customizable access policy. Link: [Create a policy](#)
 - 5. Self service:** Enable users to request access to the application using their Azure AD credentials. Link: [Get started](#)
- What's New:** This section contains three announcements, each with an icon, a title, and a description:
 - Sign in charts have moved!** The new Insights view shows sign in info along with other useful application data. Link: [View insights](#)
 - Delete Application has moved to Properties** You can now delete your application from the Properties page. Link: [View properties](#)
 - Getting started has moved to Overview** The Getting Started page has been replaced by the steps above

10. Click on Assign users and groups

[Home](#) > [Default Directory](#) > [Enterprise applications](#) > [Browse Azure AD Gallery \(Preview\)](#) > [SAP NetWeaver](#)



SAP NetWeaver | Overview

Enterprise Application

<<



Overview



Deployment Plan



Diagnose and solve problems

Manage



Properties



Owners



Roles and administrators (Preview)



Users and groups



Single sign-on



Provisioning



Self-service

Security



Conditional Access

Properties



Name ⓘ

SAP NetWeaver

Application ID ⓘ

[Redacted]

Object ID ⓘ

[Redacted]

Getting Started



1. Assign users and groups

Provide specific users and groups access to the applications

[Assign users and groups](#)

11. Click on + Add user to assign a user to this application

[Home](#) > [Default Directory](#) > [Enterprise applications](#) > [SAP NetWeaver](#)

SAP NetWeaver | Users and groups

Enterprise Application

-  Overview
-  Deployment Plan
-  Diagnose and solve problems

Manage

-  Properties
-  Owners
-  Roles and administrators (Preview)
-  **Users and groups**
-  Single sign-on
-  Provisioning
-  Self-service

Security

-  Conditional Access
-  Permissions
-  Token encryption

Activity

-  Sign-ins
-  Usage & insights (Preview)
-  Audit logs
-  Provisioning logs (Preview)
-  Access reviews

Troubleshooting + Support

-  Virtual assistant (Preview)
-  New support request

<<

+ Add user

 Edit

 Remove

 Update Credentials

 Columns

 Got feedback?



Application will appear on the Access Panel for assigned users. Set 'visible to users?' to no in properties to prevent this. →

First 100 shown, to search all users & groups, enter a display name.

Display Name

No application assignments found

12. After adding the user, we should see the user listed as below. For example, in my case it is my user ID

SAP NetWeaver | Users and groups
Enterprise Application

« + Add user Edit Remove Update Credentials Columns Got feedback?

The application will appear on the Access Panel for assigned users. Set "visible to users?" to no in properties to prevent this. →

First 100 shown, to search all users & groups, enter a display name.

Display Name	Object Type
☐ Ali Chalhouh	User

Manage

- Overview
- Deployment Plan
- Diagnose and solve problems
- Properties
- Owners
- Roles and administrators (Preview)
- Users and groups**
- Single sign-on
- Provisioning
- Self-service

Security

- Conditional Access
- Permissions
- Token encryption

Activity

- Sign-ins
- Usage & insights (Preview)
- Audit logs
- Provisioning logs (Preview)
- Access reviews

Troubleshooting + Support

- Virtual assistant (Preview)
- New support request

13. Click on Single sign-on so we can enable SAML

[Home](#) > [Default Directory](#) > [Enterprise applications](#) > [Browse Azure AD Gallery \(Preview\)](#) > [SAP NetWeaver](#)

SAP NetWeaver | Single sign-on
Enterprise Application

« Select a single sign-on method [Help me decide](#)

Disabled

Single sign-on is not enabled. The user won't be able to launch the app from My Apps.

SAML

Rich and secure authentication to applications using the SAML (Security Assertion Markup Language) protocol.

Linked

Link to an application in the Azure Active Directory Access Panel and/or Office 365 application launcher.

Manage

- Overview
- Deployment Plan
- Diagnose and solve problems
- Properties
- Owners
- Roles and administrators (Preview)
- Users and groups
- Single sign-on**
- Provisioning
- Self-service

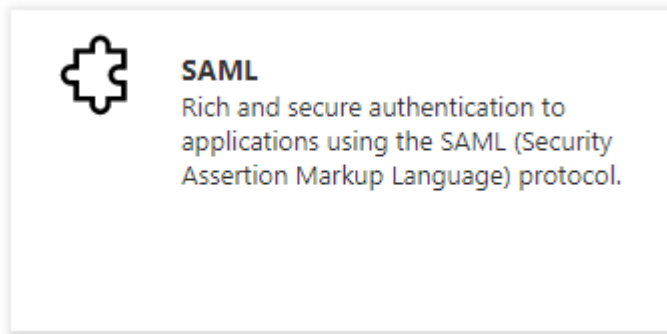
Security

- Conditional Access
- Permissions
- Token encryption

Activity

- Sign-ins

14. Click on the SAML box



15. Once clicking on SAML, we should see the screen below:

Home > Default Directory > Enterprise applications > Browse Azure AD Gallery (Preview) > SAP NetWeaver >

SAP NetWeaver | SAML-based Sign-on

Enterprise Application

« Upload metadata file Change single sign-on mode Test this application Got feedback?

- Overview
- Deployment Plan
- Diagnose and solve problems
- Manage
 - Properties
 - Owners
 - Roles and administrators (Preview)
 - Users and groups
 - Single sign-on**
 - Provisioning
 - Self-service
- Security
 - Conditional Access
 - Permissions
 - Token encryption
- Activity
 - Sign-ins
 - Usage & insights (Preview)
 - Audit logs
 - Provisioning logs (Preview)
 - Access reviews
- Troubleshooting + Support
 - Virtual assistant (Preview)
 - New support request

Set up Single Sign-On with SAML

Read the [configuration guide](#) for help integrating SAP NetWeaver.

- Basic SAML Configuration** [Edit](#)

Identifier (Entity ID)	Required
Reply URL (Assertion Consumer Service URL)	Required
Sign on URL	Required
Relay State	Optional
Logout Url	Optional
- User Attributes & Claims** [Edit](#)

givenname	user.givenname
surname	user.surname
emailaddress	user.mail
name	user.userprincipalname
Unique User Identifier	user.userprincipalname
- SAML Signing Certificate**

No certificates exist for this application. [Add a certificate](#)
- Set up SAP NetWeaver**

You'll need to configure the application to link with Azure AD.

Login URL	https://login.microsoftonline.com/c0e14236-9b8f-...
Azure AD Identifier	https://sts.windows.net/c0e14236-9b8f-4a5b-955...
Logout URL	https://login.microsoftonline.com/common/wsfed...

[View step-by-step instructions](#)
- Test single sign-on with SAP NetWeaver**

Test to see if single sign-on is working. Users will need to be added to Users and groups before they can sign in.

[Test](#)

-
16. In this step we need to upload the metadata.xml from the service provider, S/4HANA system. In this example we called the xml file s4hana_host.xml or whatever name you called the xml file when it was

downloaded. Click on the **Upload metadata file** as shown below

[Home](#) > [Default Directory](#) > [Enterprise applications](#) > [Browse Azure AD Gallery \(Preview\)](#) > [SAP NetWeaver](#) >

SAP NetWeaver | SAML-based Sign-on

Enterprise Application

- Overview
- Deployment Plan
- Diagnose and solve problems
- Manage
 - Properties
 - Owners
 - Roles and administrators (Preview)
 - Users and groups
 - Single sign-on**
 - Provisioning
 - Self-service
- Security
 - Conditional Access
 - Permissions
 - Token encryption
- Activity
 - Sign-ins
 - Usage & insights (Preview)
 - Audit logs
 - Provisioning logs (Preview)
 - Access reviews
- Troubleshooting + Support
 - Virtual assistant (Preview)
 - New support request

« [↑ Upload metadata file](#) [↶ Change single sign-on mode](#) [☰ Test this application](#) [♥ Got feedback?](#)

Set up Single Sign-On with SAML

Read the [connection guide](#) for help integrating SAP NetWeaver.

- Basic SAML Configuration** [Edit](#)

Identifier (Entity ID)	Required
Reply URL (Assertion Consumer Service URL)	Required
Sign on URL	Required
Relay State	Optional
Logout Url	Optional
- User Attributes & Claims** [Edit](#)

givenname	user.givenname
surname	user.surname
emailaddress	user.mail
name	user.userprincipalname
Unique User Identifier	user.userprincipalname
- SAML Signing Certificate**

No certificates exist for this application. [Add a certificate](#)
- Set up SAP NetWeaver**

You'll need to configure the application to link with Azure AD.

Login URL	https://login.microsoftonline.com/c0e14236-9b8f-...
Azure AD Identifier	https://sts.windows.net/c0e14236-9b8f-4a5b-955...
Logout URL	https://login.microsoftonline.com/common/wsfed...

[View step-by-step instructions](#)
- Test single sign-on with SAP NetWeaver**

Test to see if single sign-on is working. Users will need to be added to Users and groups before they can sign in.

[Test](#)

17. Select the metadata xml file and click on Add

Upload metadata file.

Values for the fields below are provided by SAP NetWeaver. You may either enter those values manually, or upload a pre-configured SAML metadata file if provided by SAP NetWeaver.

Select a file

[Add](#) [Cancel](#)

18. We should see some like that below where the Identifier and Reply URL are being populated

Overview

Deployment Plan

Diagnose and solve problems

Manage

Properties

Owners

Roles and administrators (Preview)

Users and groups

Single sign-on

Provisioning

Self-service

Security

Conditional Access

Permissions

Token encryption

Set up Single Sign-On with SAML

Read the [configuration guide](#) for help integrating SAP NetWeaver.

1 Basic SAML Configuration [Edit](#)

Identifier (Entity ID)	
Reply URL (Assertion Consumer Service URL)	
Sign on URL	
Relay State	
Logout Url	

2 User Attributes & Claims [Edit](#)

givenname	user.givenname
surname	user.surname
emailaddress	user.mail
name	user.userprincipalname
Unique User Identifier	user.userprincipalname

19. Next step we need to edit the User Attributes & Claims which is required for the SAML assertions and it has to be in a specific format but for S/4HANA or NetWeaver, we need to provide a custom one. Click on

the Edit pencil as shown below

User Attributes & Claims		 Edit
givenname	user.givenname	
surname	user.surname	
emailaddress	user.mail	
name	user.userprincipalname	
Unique User Identifier	user.userprincipalname	

20. Click on Unique User Identifier (Name ID)

[Home](#) > [Default Directory](#) > [Enterprise applications](#) > [SAP NetWeaver](#)

User Attributes & Claims

[+](#) Add new claim [+](#) Add a group claim [≡](#) Columns

Required claim

Claim name

Unique User Identifier (Name ID)

Additional claim

Claim name

<http://schemas.xmlsoap.org/ws/2005/05/identity/claims/emailaddress>

<http://schemas.xmlsoap.org/ws/2005/05/identity/claims/givenname>

<http://schemas.xmlsoap.org/ws/2005/05/identity/claims/name>

<http://schemas.xmlsoap.org/ws/2005/05/identity/claims/surname>

21. From the Manage Claim click on Transformation

Manage claim

[Save](#) [X Discard changes](#)

Name	nameidentifier
Namespace	http://schemas.xmlsoap.org/ws/2005/05/identity/claims
^ Choose name identifier format	
Unspecified	
Source *	☒ Attribute ☐ Transformation
Source attribute *	user.userprincipalname
v Claim conditions	

22. Click on Transformation. We should see Undefined. Click on the Pencil to add a transformation

Manage claim

 Save  Discard changes

Name

Namespace

^ Choose name identifier format

Source * ☐ Attribute ☒ Transformation

Transformation **Undefined** 

v Claim conditions

23. We need to add a transformation from the below screen

Enterprise applications > SAP NetWeaver > SAML-based Sign-on > User Attributes & Claims >

×

User Attributes & Claims

+ Add new claim + Add a group claim Columns

Required claim

Claim name	Value
Unique User Identifier (Name ID)	user.userprincipalname [nameid]

Additional claims

Claim name	Value
http://schemas.xmlsoap.org/ws/2005/05/identity/claims/emailaddress	user.mail
http://schemas.xmlsoap.org/ws/2005/05/identity/claims/givenname	user.givenname
http://schemas.xmlsoap.org/ws/2005/05/identity/claims/name	user.userprincipalname
http://schemas.xmlsoap.org/ws/2005/05/identity/claims/surname	user.surname

Manage transformation

×

Transformation *

Select from drop down

Parameter 1 *

Select from drop down or type a value

+ Add transformation

24. By default, S/4HANA or NetWeaver expects a Logon ID to be sent in the SAML Assertion (value populated to the NameID attribute of the SAML assertion).. Therefore; we need to extract the User ID from the Microsoft email
25. To-do that, set the following transformation as shown below and click Add

Manage transformation



Transformation *	ExtractMailPrefix()
Parameter 1 *	user.mail
<button>+ Add transformation</button>	

Add



26. We should have something like this screen, click on Save

Manage claim

Save Discard changes



Name	nameidentifier
Name space	http://schemas.xmlsoap.org/ws/2005/05/identity/claims
^ Choose name identifier format	
Unspecified	
Source *	☐ Attribute ☒ Transformation
Transformation	ExtractMailPrefix (user.mail)
v Claim conditions	

i Note

If the requirement to use email ID and not Logon ID, then select Attribute instead of Transformation and select from the Source Attributer "user.mail"
In S/4HANA you need to go to SU01 and find the user that you have and the Microsoft email address to the email field and Make sure the User ID Mapping Mode for the NameID format in the Identity Federation is set to mail as shown below:

Details of NameID Format "Unspecified"

User ID Source:	Assertion Subject NameID	User ID Mapping Mode:	Email
Assertion Attribute Name:		Allow Identity Provider to Create NameID:	Yes

27. Our User Attributes & Claims should have the following configuration below:

User Attributes & Claims

[+ Add new claim](#) [+ Add a group claim](#) [☰ Columns](#)

Required claim

Claim name	Value
Unique User Identifier (Name ID)	ExtractMailPrefix (user.mail) [nameid... ***

Additional claims

Claim name	Value
http://schemas.xmlsoap.org/ws/2005/05/identity/claims/emailaddress	user.mail ***
http://schemas.xmlsoap.org/ws/2005/05/identity/claims/givenname	user.givenname ***
http://schemas.xmlsoap.org/ws/2005/05/identity/claims/name	user.userprincipalname ***
http://schemas.xmlsoap.org/ws/2005/05/identity/claims/surname	user.surname ***

28. Now we need to go back and download the Federation Metadata XML, click on SAML-based Sign-on

[Home](#) > [SAP NetWeaver](#) > [SAML-based Sign-on](#) >

User Attributes & Claims



[+ Add new claim](#) [+ Add a group claim](#) [☰ Col](#)

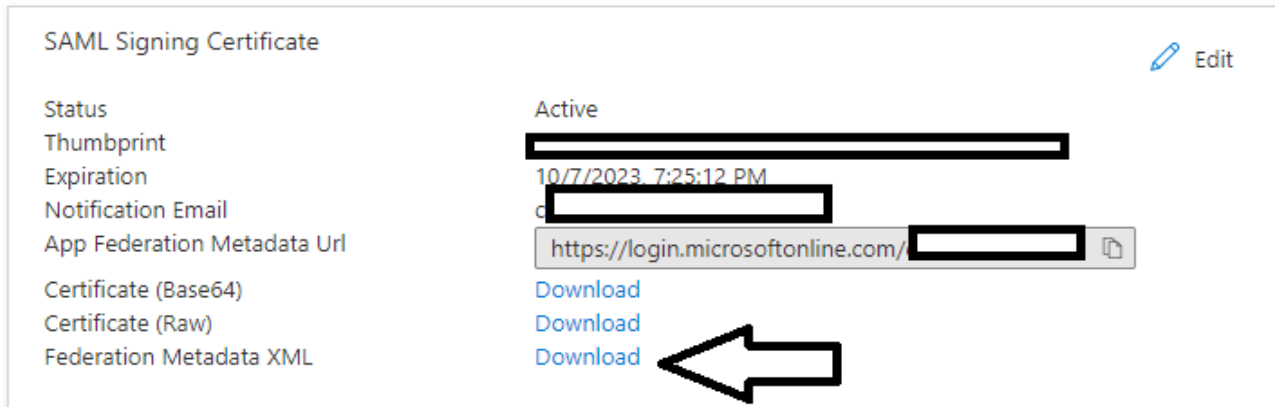
Required claim

Claim name
Unique User Identifier (Name ID)

Additional claims

Claim name
http://schemas.xmlsoap.org/ws/2005/05/identity/claim
http://schemas.xmlsoap.org/ws/2005/05/identity/claim
http://schemas.xmlsoap.org/ws/2005/05/identity/claim
http://schemas.xmlsoap.org/ws/2005/05/identity/claim

29. Click on Download as shown below

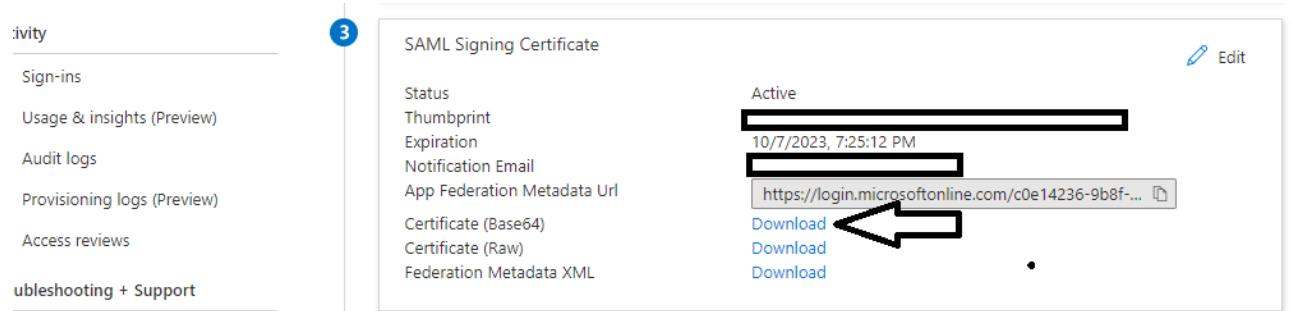


SAML Signing Certificate Edit

Status	Active
Thumbprint	[REDACTED]
Expiration	10/7/2023, 7:25:12 PM
Notification Email	[REDACTED]
App Federation Metadata Url	https://login.microsoftonline.com/[REDACTED]
Certificate (Base64)	Download
Certificate (Raw)	Download
Federation Metadata XML	Download

30. Save the file to your local drive. By default, the name is “SAP NetWeaver.xml”. You can name it anything you like or keep it as the default.

31. Next download the Azure Certificate. Click on Download beside Certificate (Base64)



3 SAML Signing Certificate Edit

Status	Active
Thumbprint	[REDACTED]
Expiration	10/7/2023, 7:25:12 PM
Notification Email	[REDACTED]
App Federation Metadata Url	https://login.microsoftonline.com/c0e14236-9b8f-...
Certificate (Base64)	Download
Certificate (Raw)	Download
Federation Metadata XML	Download

32. Save it to your local disk. By default it is called “SAP Netweaver.cer”

Importing Azure Active Directory Identity Provider Certificate into Service Provider

1. Now that the Metadata has been downloaded go back to the Service Provider and access your SAML2 configuration screen as shown below by either using tCode **saml2** or access the SAML2 configuration by using the URL. Example:

`http(s)://<HOST-NAME>:<PORT>/sap/bc/webdynpro/sap/saml2?sap-client=<CLIENT-ID>`

SAML 2.0 Configuration of ABAP System:

Local Provider **Trusted Providers** Policies Name ID Management

List of Trusted Provider 

Show: Identity Providers v | Edit Save Cancel Disable Add v Remove

	Active	Default	Name	Alias
☐				Manually
☐				Upload Metadata File 
☐				
☐				
☐				

2. Select the Azure Federation XML file

SAML 2.0 Configuration

New Trusted Identity Provider

1 Select Metadata 2 Metadata Verification 3 Select Providers 4 Provider Name 5 Signature and Encryption 6 Single Sign-On Endpoints 7 Single Logout Endpoints →

< Previous Next > Finish Cancel

* Metadata File: SAP NetWeaver.xml Browse...

3. Click on Next
4. We should see the following screen below:

New Trusted Identity Provider

1 Select Metadata 2 Metadata Verification 3 Select Providers 4 Provider Name 5 Signature and Encryption 6 Single Sign-On Endpoints 7 Single Logout Endpoints 8 Artifact Endpoints 9 Authentication Requirements

< Previous Next > Finish Cancel

Metadata is signed with the following certificate: CN=Microsoft Azure Federated SSO Certificate. Specify local copy of the signing certificate to ensure trust

☒ Search in Address Book

Subject: 🔗

Issuer:

☐ Upload from File

File: Browse...

< Previous Next > Finish Cancel

5. We need to provide now the certificate of the IDP which is Azure Active Directory Single Sign-on. Click on Upload from File and select browse to select "SAP Netweaver.cer" or whatever the file name is called and click on Next

6. Provide an alias

New Trusted Identity Provider

1 — 2 — 3 — 4 — 5

Select Metadata Metadata Verification Select Providers **Provider Name** Signature and

< Previous Next > Finish Cancel

* Name:

Alias:

7. Click on Next
8. We should see the following screen below. Note: Make sure Digest Algorithm is set to SH-256

New Trusted Identity Provider

1 — 2 — 3 — 4 — 5 — 6 — 7

Select Metadata Metadata Verification Select Providers Provider Name **Signature and Encryption** Single Sign-On Endpoints Single Logout Endpoints

< Previous Next > Finish Cancel

Certificates and Algorithms

* Primary Signing Certificate: [Details](#) [Browse](#)

Secondary Signing Certificate: [Details](#) [Browse](#)

Digest Algorithm: ▼

Encryption Certificate: [Details](#) [Browse](#)

Single Sign-On Authentication Request

Sign: ▼

Single Sign-On Assertions

Require Signature: ▼

Single Sign-On Response

Require Signature: ▼

Require Encrypted Elements: ▼

Logout Request and Logout Response

Sign: ▼

Require Signature: ▼

Encrypt elements: ▼

Require Encrypted Elements: ▼

Artifact Profile

Sign: ▼

Require Signature: ▼

< Previous Next > Finish Cancel

9. Click on Next. Now we are on the Single Sign-on Endpoints

New Trusted Identity Provider

1 Select Metadata 2 Metadata Verification 3 Select Providers 4 Provider Name 5 Signature and Encryption 6 **Single Sign-On Endpoints** 7 Single Logout Endpoints 8 Artifact Endpoints 9 Authentication Requirements

< Previous Next > Finish Cancel

Single Sign-On Endpoints

Add Remove

* Default	Binding	Location URL
☒	HTTP Redirect	https://login.microsoftonline.com/c0[redacted]
☐	HTTP POST	https://login.microsoftonline.com/c0[redacted]
☐		
☐		
☐		
☐		
☐		
☐		
☐		

< Previous Next > Finish Cancel

10. Click on Next

New Trusted Identity Provider

1 Select Metadata 2 Metadata Verification 3 Select Providers 4 Provider Name 5 Signature and Encryption 6 Single Sign-On Endpoints 7 **Single Logout Endpoints** 8 Artifact Endpoints 9 Authentication Requirements

< Previous Next > Finish Cancel

Single Logout Endpoints

Add Remove

* Default	Binding	Location URL
☒	HTTP Redirect	https://login.microsoftonline.com/c0e[redacted]
☐		
☐		
☐		
☐		
☐		
☐		
☐		
☐		

< Previous Next > Finish Cancel

11. Click on Next

12. Click on Edit

SAML 2.0 Configuration of ABAP System: [redacted]

Local Provider Trusted Providers Policies Name ID Management

List of Trusted Providers

Show: Identity Providers Edit Save Cancel Enable Add Remove

Active	Default	Name	Alias
☒	☒	https://sts.windows.net/c0[redacted]	[redacted]
☐			
☐			
☐			

Details of Identity Provider "https://sts.windows.net/[redacted]"

Endpoints Identity Federation Signature and Encryption Authentication Requirements

Supported NameID Formats

Add Remove

Default	Name
---------	------

13. Click on Add as shown below to add a NameID attribute

SAML 2.0 Configuration of ABAP System: [redacted]

Local Provider **Trusted Providers** Policies Name ID Management

List of Trusted Providers

Show: Identity Providers Edit Save Cancel [redacted] Add v Remove

	Active	Default	Name	Alias
☒	☒	☒	https://sts.windows.net/c0[redacted]	[redacted]
☐	☐	☐		
☐	☐	☐		
☐	☐	☐		
☐	☐	☐		

Details of Identity Provider "https://sts.windows.net/c0[redacted]"

Endpoints **Identity Federation** Signature and Encryption Authentication Requirements

Supported NameID Formats

Add Remove

	Default	Name
☒		
☐		
☐		
☐		
☐		
☐		

14. Select "Unspecified"

SAML 2.0 Configuration of ABAP System: [Redacted]

Local Provider **Trusted Providers** Policies Name ID Management

List of Trusted Providers

Show: Identity Providers | Edit | Save | Cancel | Enable | Add v | Remove

	Active	Default	Name
☒	◆	☒	https://sts.windows.net/c0a[Redacted]
☐			
☐			
☐			
☐			

Details of Identity Provider "https://sts.windows.net/c0a[Redacted]"

Endpoints **Identity Federation** Signature and Encryption Authentication Requirements

Supported NameID Formats

Add Remove

	Default	Name
☐		
☐		
☐		
☐		
☐		
☐		
☐		

Supported NameID For... [Link Icon]

	Name
☒	Unspecified
☐	X509 Subject Name
☐	Windows Name
☐	Kerberos
☐	Transient
☐	Persistent
☐	E-mail

OK Cancel

15. Make sure User ID Mapping Mode is set to Logon ID as shown below. If all OK, click on Save

Local Provider

Trusted Providers

Policies

Name ID Management

List of Trusted Providers

Show: Identity Providers

Edit

Save

Cancel

Enable

Add

Remove

	Active	Default	Name
☒	☒	☒	https://sts.windows.net/c0e[REDACTED]
☐	☐	☐	
☐	☐	☐	
☐	☐	☐	
☐	☐	☐	

Details of Identity Provider "https://sts.windows.net/c0e[REDACTED]"

Endpoints

Identity Federation

Signature and Encryption

Authentication Requirements

Supported NameID Formats

Add

Remove

	Default	Name
☒	☒	Unspecified
☐	☐	
☐	☐	
☐	☐	
☐	☐	
☐	☐	

Details of NameID Format "Unspecified"

User ID Source: Assertion Subject NameID

User ID Mapping Mode: Logon ID

Assertion Attribute Name:

Allow Identity Provider to Create NameID: Yes

16. Finally, we need to enable our trusted provider. This is very important because if we do not, authentication with Azure will not take place. Click on Enable

Local Provider Trusted Providers Policies Name ID Management

List of Trusted Providers

Show: Identity Providers

	Active	Default	Name
☒	☒	☒	https://sts.windows.net/c0e14236-9b8f-4a5b-955e-ea98a53deb0c/
☐	☐	☐	
☐	☐	☐	
☐	☐	☐	
☐	☐	☐	

Details of Identity Provider "https://sts.windows.net/c0e14236-9b8f-4a5b-955e-ea98a53deb0c/"

Endpoints Identity Federation Signature and Encryption Authentication Requirements

Supported NameID Formats

	Default	Name
☒	☒	Unspecified
☐	☐	
☐	☐	
☐	☐	
☐	☐	
☐	☐	

Details of NameID Format "Unspecified"

User ID Source: User ID Mapping Mode:

Assertion Attribute Name: Allow Identity Provider to Create NameID:

17. We should see this confirmation. Click on OK

SAML 2.0 Configuration

Are you sure you want to enable trusted provider "https://sts.windows.net/c0e14236-9b8f-4a5b-955e-ea98a53deb0c/"?

18. Next, we need configure the relay state, click on **Local Provider** Tab

SAML 2.0 Configuration of ABAP System:

Local Provider Trusted Providers Policies Name ID Management

Provider Name:

Operation Mode:

Status: ☒ Enabled

General Settings Authentication Contexts Service Provider Settings

19. Click on Service Provider Settings

SAML 2.0 Configuration of ABAP System:

Local Provider Trusted Providers Policies Name ID Management

Provider Name:

Operation Mode:

Status: ☒ Enabled

General Settings Authentication Contexts Service Provider Settings

20. Click on Edit button

21. Scroll down until you see Relay State Mapping. Click on Add as shown below:

Miscellaneous

Affiliation Name:

Legacy Systems Support (Issue Logon Ticket):

Off

Enable Debugging:

No

Assertion Consumer Service

Supported Bindings: ☒ HTTP POST ☒ HTTP Artifact ☒ PAOS

Endpoint Path: /sap/saml2/sp/acs/100

Default Application Path:

RelayState Mapping

Add	Remove
☒ * RelayState	☐ * Application Path
☐	
☐	

22. Enter a Relay State name and the Fiori launchpad path as shown below:

SAML 2.0 Configuration

* RelayState: fiori

* Path: /sap/bc/ui5_ui5/ui2/ushell/shells/abap/FioriLaunchpad.html

OK Cancel

RelayState: fiori

Path: /sap/bc/ui5_ui5/ui2/ushell/shells/abap/FioriLaunchpad.html

23. Click on OK

24. Repeat the process and add the following:

a. RelayState: it00

b. Path: /sap/bc/bsp/sap/it00/default.htm

25. Click on OK

26. We should have the following:

Assertion Consumer Service

Supported Bindings: ☒ HTTP POST ☒ HTTP Artifact ☒ PAOS

Endpoint Path: /sap/saml2/sp/acs/

Default Application Path:

RelayState Mapping

Add Remove

☒ * RelayState	* Application Path
☒ fiori	/sap/bc/ui5_ui5/ui2/ushell/shells/abap/FioriLaunchpad.html
☒ it00	/sap/bc/bsp/sap/it00/default.htm
☐	
☐	

 **Note**

Make sure the service /sap/saml2/sp/acs/<CLIENT> is activated.

27. Click on Save

Configuring Fiori Launchpad to Support SAML2

Configuring Fiori Launchpad

In this section Fiori launchpad needs to be configured to support SAML2. In this section, we will go through all the steps needed to allow Fiori launchpad to support SAML2 authentication.

1. Login to the S/4HANA System
2. Execute tCode SICF
3. Under Service Name type **USHELL**

The screenshot displays the 'Define Services' SAP transaction interface. It is divided into two main sections: 'Filter for Calling ICF Hierarchy' and 'Filter for Detail Information'.

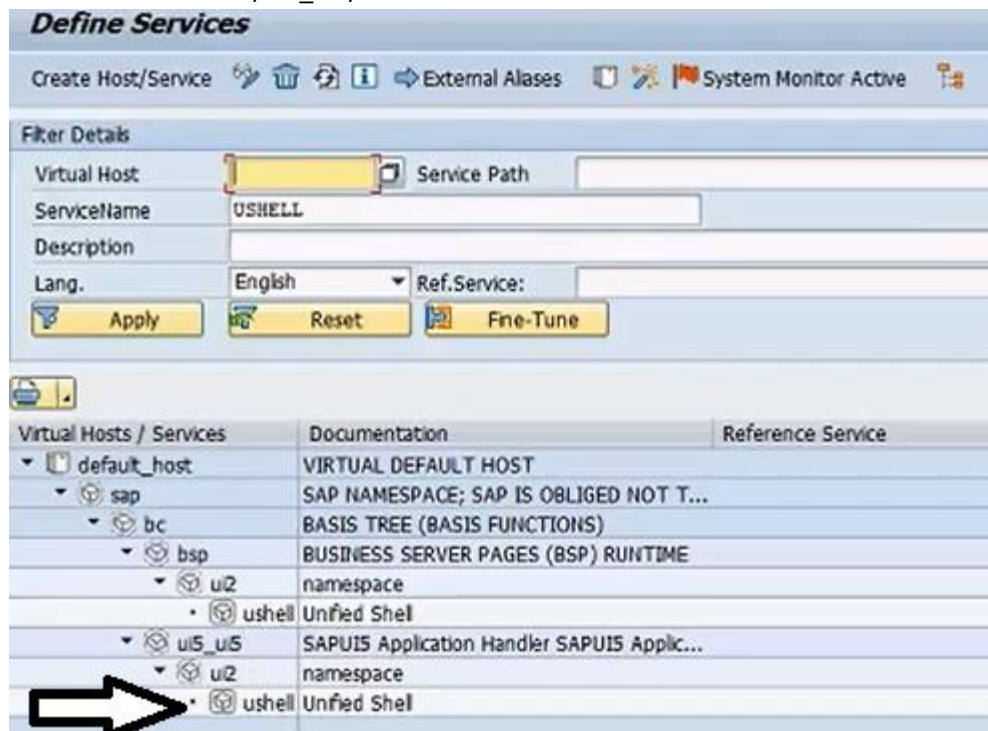
Filter for Calling ICF Hierarchy:

Hierarchy Type	SERVICE
Virtual Host	
Service Path	
Service Name	ushell
Reference Service	
Description	
Language	English

Filter for Detail Information:

Created By		to	
Created On		to	
Last Changed By		to	
Changed On		to	

4. Press F8 to execute
5. Click on **ushell** under /ui5_ui5/ui2 as shown below



6. Click on Logon Data tab. In here there two options that the admin can follow. You just need one of those options and not both

- a. First option is to use under Procedure “Standard

The screenshot shows a configuration page for a service. At the top, the Path is set to '/default_host/sap/bc/ui5_ui5/ui2/'. Below this, the Service Name is 'ushe11' and the Language is 'English'. A section titled 'Description' contains three text boxes: 'Description 1: Unified Shell', 'Description 2:', and 'Description 3:'. Below the description section is a tabbed interface with five tabs: 'Service Data', 'Logon Data', 'Handler List', 'Error Pages', and 'Admin'. The 'Logon Data' tab is selected. In this tab, the 'Procedure' is set to 'Standard'. Below the procedure dropdown, there is a checkbox labeled 'Use All Logon Procedures' which is checked. Two large black arrows with orange numbers are overlaid on the image: arrow '1' points to the 'Logon Data' tab, and arrow '2' points to the 'Use All Logon Procedures' checkbox.

Or

- b. Second option to explicitly indicate SAML is used and follow the steps below. As mentioned, you need either a) or b)

Create/Change a Service

Path: /default_host/sap/bc/u05_u05/u02/

Service Name: ushell Service (Active)

Lang.: English Other Languages

Description

Description 1: Unified Shell

Description 2:

Description 3:

Service Data Logon Data Handler List Error Pages Administration

Procedure: Alternative Logon Procedure SAML Configuration

☐ Use All Logon Procedures Security Session: Unrestricted

Logon Data

Client:

User:

Language:

Password Status: Initial

Security Requirement

☒ Standard ☐ SSL

Authentication

☒ Standard SAP User ☐ Internet User

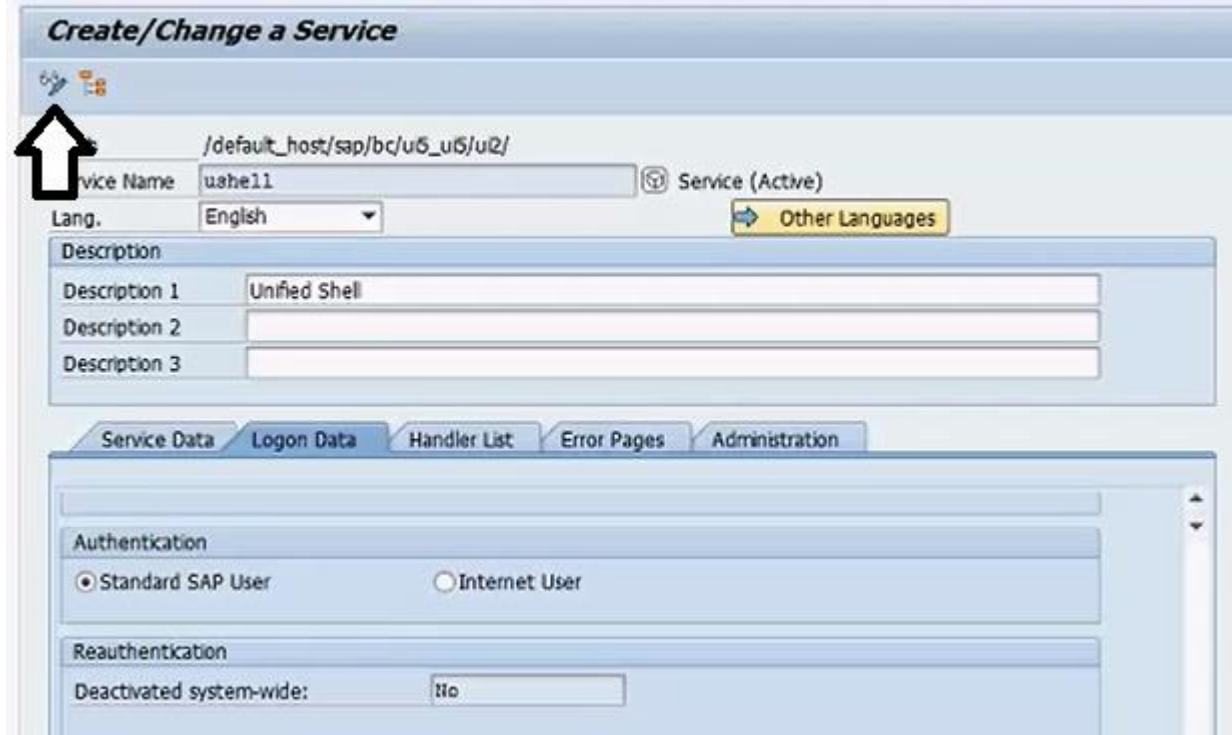
Reauthentication

Deactivated system-wide: No

Default

7. If you Procedure is set to "Standard" and "Use All Logon Procedures" is ticked, you can skip this section. And if Alternative Logon Procedure and SAML Configuration already set, then you are done on this section. If not, then follow these steps below:

- a. Click on Edit



Create/Change a Service

  /default_host/sap/bc/u5_u5/u2/

Service Name  Service (Active)

Lang. 

Description

Description 1

Description 2

Description 3

Service Data Logon Data Handler List Error Pages Administration

Authentication

☒ Standard SAP User ☐ Internet User

Reauthentication

Deactivated system-wide:

- b. Under Procedure drop down list change it from **Standard** to **Alternative Logon Procedure**
- c. In the Logon Data section scroll down

- d. Change the Logon Procedure List by scrolling all the way until 8 SAML Logon is shown

Create/Change a Service

Path: /default_host/sap/bc/u05_u05/u02/

Service Name: ushell Service (Active)

Lang.: English Other Languages

Description

Description 1: Unified Shell

Description 2:

Description 3:

Service Data Logon Data Handler List Error Pages Administration

Authentication

☒ Standard SAP User ☐ Internet User

Reauthentication

Deactivated system-wide: No

Default

Logon Procedure List (in Order of Execution)

N.	Logon Procedure
8	SAML Logon
9	Logon Through Service Data

- e. Change 8 to 1

Default

Logon Procedure List (in Order of Execution)

N.	Logon Procedure
1	SAML Logon
9	Logon Through Service Data

- f. Press Enter

g. We should see the following result

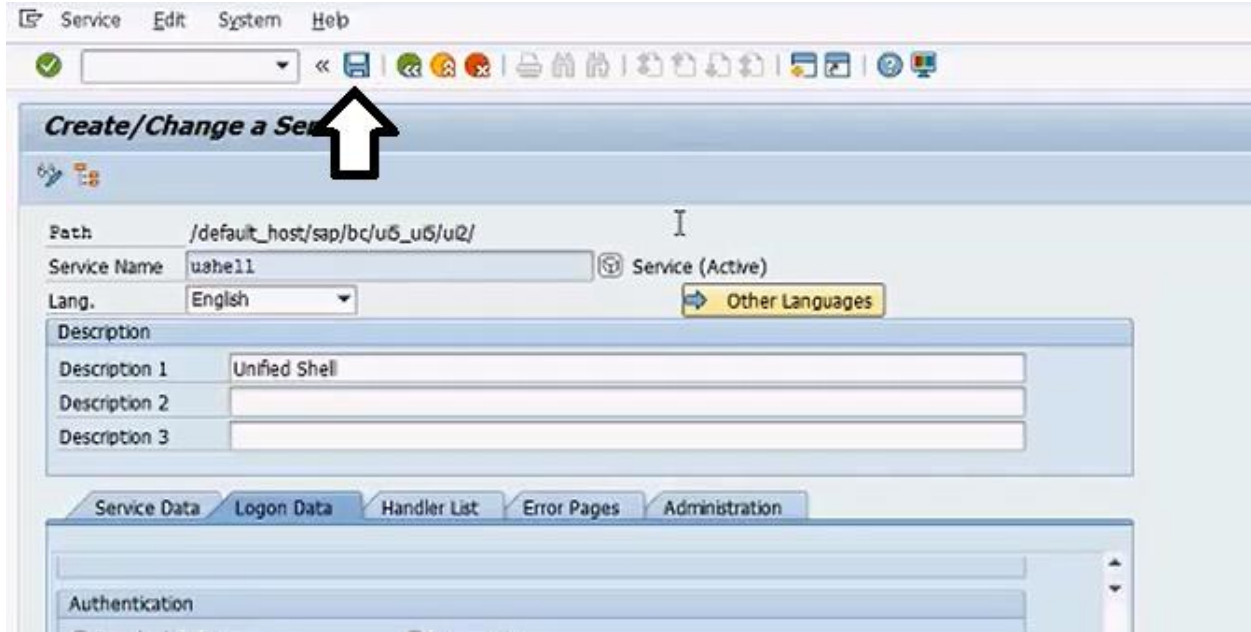
The screenshot shows the SAP NetWeaver Administration console with the 'Logon Data' tab selected. The 'Authentication' section has 'Standard SAP User' selected. The 'Reauthentication' section shows 'Deactivated system-wide' set to 'No'. Below, the 'Logon Procedure List (in Order of Execution)' is displayed for the 'Default' profile. The list contains five entries: 1. Logon Through HTTP Fields, 2. SAML Logon, 3. Logon Through SSL Certificate, 4. SAP Logon/Assertion Ticket, and 5. SAP Assertion Ticket. A white arrow points to the 'SAML Logon' entry, which is at position 2.

N.	Logon Procedure
1	Logon Through HTTP Fields
2	SAML Logon
3	Logon Through SSL Certificate
4	SAP Logon/Assertion Ticket
5	SAP Assertion Ticket

Note

Even though we set the order to be 1, Logon Through HTTP Fields is always 1 and then comes our SAML Logon based on the order we set.

h. Click on Save



Testing SAML Using Fiori launchpad

To test the configuration, we need to access Fiori launchpad

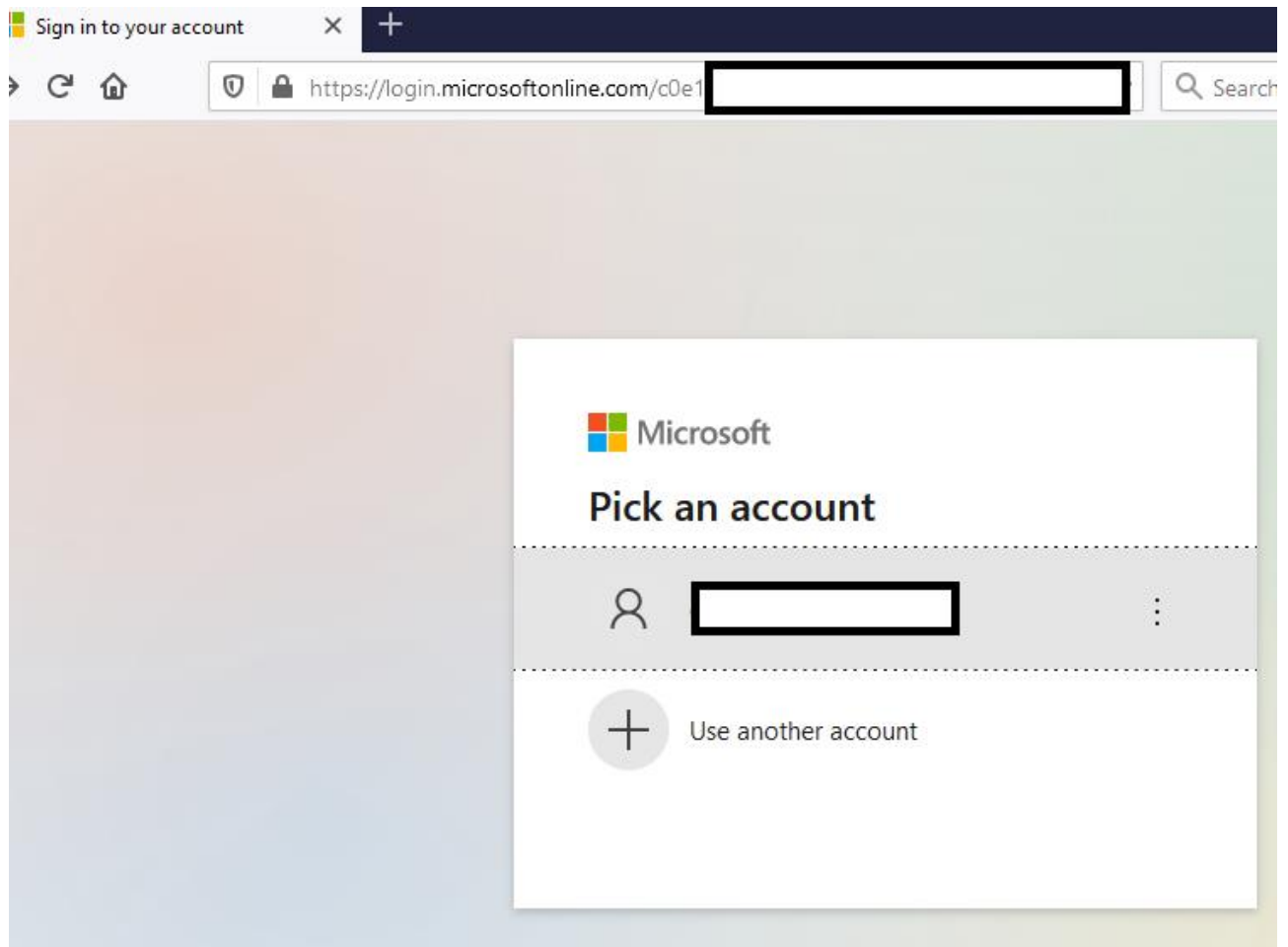
1. Open web browser, preferably Chrome
2. Enter the URL of your Fiori launchpad

http://<DOMAIN>:8443/sap/bc/ui5_ui5/ui2/ushell/shells/abap/FioriLaunchpad.html

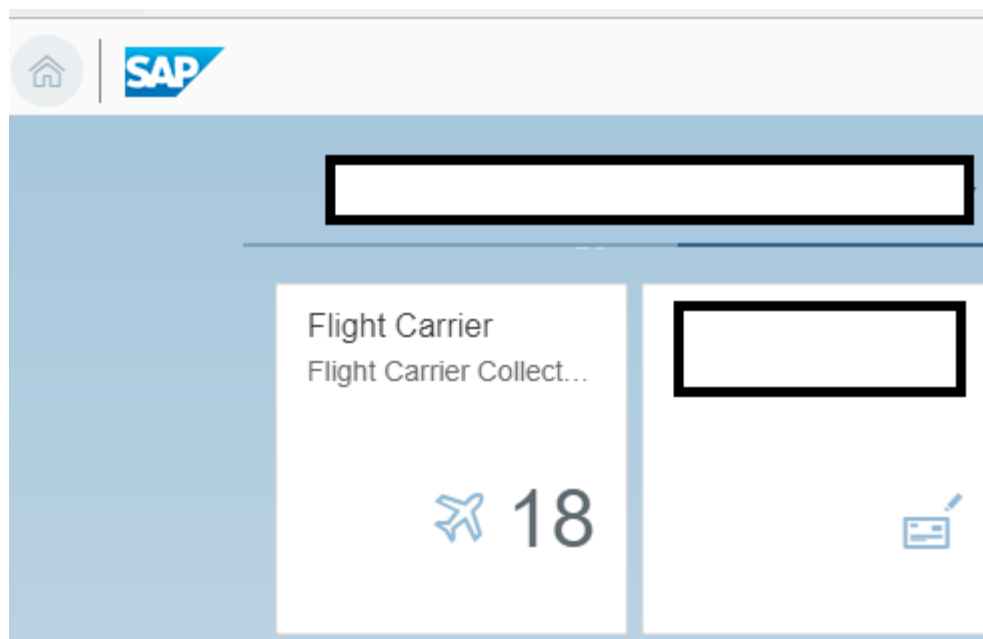
i Note

Because we configured our Service Provider by going to the following URL <http://<DOMAIN>/sap/bc/webdynpro/sap/saml2?sap-client=<CLIENT-ID>&sap-language=EN#>. WE MUST access Fiori launchpad using http and not HTTPS. We will see how we can change this later.

3. If everything is configured correctly, the web browser will redirect the request to the Microsoft Azure IDP as shown below:



4. Login with your IDP user ID and password. Fiori launchpad should log you in successfully



Configuring Fiori Launchpad Designer

In this section we will configure Fiori Launchpad Designer to support SAML2.

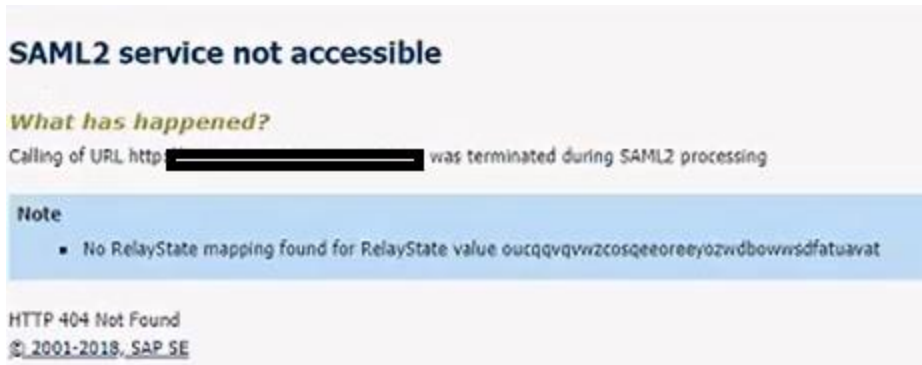
1. Login into S/4HANA or Netweaver Gateway system
2. Execute tCode **/nSAML2**
3. Click on **Local Provider**
4. Click on **Service Provider Settings**
5. Click on **Edit** button on the top left
6. Under Relay State Mapping click on **Add**
7. Enter the following configuration:
RelayState: **fioridesigner**
Path: **/sap/bc/ui5_ui5/sap/arsvc_upb_admn/main.html**
8. Click **OK**
9. Click **Save**
10. Configuration should look like this screen below:

RelayState Mapping	
Add	Remove
*RelayState	*Application Path
fiori	/sap/bc/ui5_ui5/ui2/ushell/shells/abap/FioriLaunchpad.html
fioridesigner	/sap/bc/ui5_ui5/sap/arsvc_upb_admn/main.html

Chapter 3

Troubleshooting

Error 1 – No RelayState mapping found for RelayState value



Solution:

No RelayState has been created. To fix this issue, follow the steps under Chapter1 section that talks about relaystate.

How to trace SAML issues in S/4HANA or Netweaver

To trace SAML, please follow the following KBA: [2501320](#) - How to get necessary traces for analyzing SAML2 issue in Netweaver ABAP system

Microsoft Tutorial on SAML and Netweaver using Azure

For more information about configuration of SAML and Netweaver using Azure from Microsoft, please refer to this URL: <https://docs.microsoft.com/en-us/azure/active-directory/saas-apps/sap-netweaver-tutorial>